

BookBorrow

Peer-to-Peer Book Lending Platform

Project Progress & Handover Report

CITS5206 Software Engineering Capstone | University of Western Australia

Development Period: March – May 2026

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1. Executive Summary

BookBorrow is a peer-to-peer book lending platform developed as the CITS5206 Software Engineering Capstone project at the University of Western Australia. The platform enables users to list books for lending, browse and borrow available titles, manage shipments, and process payments and deposits — all within a secure, role-based web application.

This report consolidates all progress updates and meeting outcomes from Weeks 5 through 11 (March – May 2026), covering the full Phase 2 development cycle. It is intended as both a project summary for the client and a handover reference for future maintainers.

The platform is built on the following technology stack:

- Frontend: Next.js
 - Backend: FastAPI (Python)
 - Database: MySQL
 - Infrastructure: Docker, Nginx, VPS
 - Payments: Stripe (sandbox → production)
 - Email: Brevo (sender domain: noreply@bookborrow.org)
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2. Development Progress by Sprint

Sprint 1 — Week ending 31 March 2026

Focus: Environment stabilisation, foundational features, and initial UI drafts.

Dependency & Environment Update

- Audited and updated all outdated backend dependencies using pip list —outdated.
- Resolved a bcrypt version mismatch (standardised to 4.3.0) that caused a login failure after the upgrade.
- Updated all frontend packages via npx npm-check-updates -u && npm install.

Email & Domain Setup

- Purchased an email domain and configured it with Brevo to handle outgoing system emails.
- Implemented OTP-based registration email confirmation.
- Implemented forgot / reset password emails.

Forgot Password Feature

- Reactivated the Forgot Password function on the login page (previously commented out by the Phase 1 group).
- Full end-to-end flow: user requests reset → receives email → sets new password.

UI Drafts (Frontend only)

- Refund status notification page — UI draft completed; backend not yet connected.
- Admin analytics dashboard page — UI draft completed; backend not yet connected.

Sprint 2 — Week ending 14 April 2026

Focus: Email infrastructure migration, transaction flow, admin frontend foundations, and security hardening.

Brevo & Email Configuration Migration

- Created a new Brevo account to replace the one previously tied to Jinho's personal Google account, giving the team full ownership.
- Replaced existing DNS records on Namecheap with newly generated Brevo records.
- Authenticated bookborrow.org as the verified sender domain.
- Generated a new API key and updated the .env file accordingly.
- Email functionality (OTP, notifications, password reset) is fully operational under the team's account.

Change Password Feature

- Added Change Password functionality to the user profile edit page.

Transaction Flow Improvements

- Strengthened checkout and payment result flows with clearer status handling and improved error feedback.
- Improved transaction status visibility to better reflect the full lifecycle: pending → paid → shipped.
- Buyers receive an email notification containing the tracking number once sellers dispatch their shipment.

Admin Frontend Development

- Expanded admin dashboard with panels for user management and complaint management.

Responsive Design

- Improved mobile responsiveness across key pages.
- Refactored chat / messaging layout for better small-screen usability.

Security Fix

- Fixed a critical bug where users could be incorrectly authenticated as admin if their username or email contained the word "admin".
- Admin authentication across all pages now relies solely on a dedicated boolean flag returned by the backend, eliminating string-matching vulnerabilities.

Sprint 3 — Week ending 21 April 2026

Focus: Responsive design fixes, admin dashboard expansion, and checkout fee clarifications.

Responsive Design Fixes

- Book listing filter panel was hidden on mobile; now displays correctly on all screen sizes.

- Fixed “Share My Book” button that was non-functional on mobile.

Admin Dashboard Enhancements

- Enhanced User Metrics, Book Metrics, and Order Metrics pages.
- Designed and developed a Financial Metrics page providing platform-level financial insights:
 - Platform fees collected
 - Total transaction volume
 - Refund rate
 - Payment distribution breakdown
- Developed Top Earning Users feature based on lender earnings.
- Implemented a Recent Transactions view for real-time activity monitoring.

Checkout & Fee Clarity

- Added automatic rental fee calculation at checkout ($\text{rental/day} \times \text{rental days}$).
- Added deposit fee notation and tooltip on the lender listing page to clearly explain how the deposit is held.

Sprint 4 — Week ending 28 April 2026

Focus: Admin dashboard completion, checkout flow refinement, and initial VPS deployment.

Admin Dashboard — Order & Shipping Management

- Implemented Order Management: admins can view order owner, borrower, book, due date, and payment details.
- Implemented Platform Fee Setting: admins can update the transaction fee dynamically without touching the codebase (stored in DB, fetched via API).
- Created a Shipping Dashboard tracking:
 - Delivery and pickup orders
 - Orders missing tracking numbers
 - Outbound and return tracking
 - Shipping fees and recent shipments
- Connected all admin frontend panels to FastAPI backend endpoints.
- Protected all sensitive dashboard pages with admin-only access checks.
- Improved admin UX with KPI cards, filterable tables, status badges, loading states, and error handling.

Checkout Flow Refinement

- Replaced fixed lending duration input with a rental/day slider (range: \$0.00 – \$2.00/day).
- Added tooltip for Rental/Day field, including support for \$0/day listings.
- Added automatic rental fee calculation: $\text{rental/day} \times \text{rental days}$.
- Changed service fee from a percentage to a fixed \$2 per transaction.
- Separated rental details from delivery details in the checkout UI.
- Shipping fee now auto-calculates after delivery method selection.
- Improved shipping action labels (shipped orders show “Update Shipping”).

- Prevented users from requesting their own listed books.
- Unified backend datetime serialisation to UTC ISO format; unified frontend datetime formatting across borrowing, messaging, and refunds.

VPS Deployment

- Received full VPS access from Jinho.
- Successfully deployed the repository to the VPS server for testing.

Sprint 5 — Week ending 5 May 2026

Focus: Secure payment flow implementation, deposit arbitration, and production deployment.

Secure Payment Flow

A complete, secure end-to-end payment lifecycle was implemented:

- Borrower pays → full amount (rental fee + deposit + shipping) is held on the platform account.
- Lender confirms shipment → platform transfers rental fee + shipping fee to the lender's connected Stripe account; deposit remains held.
- Lender reports damage + admin confirms deduction → deducted portion is transferred to lender as compensation.
- Borrower claims refund → remaining deposit is refunded to the borrower's original payment method via Stripe.

Lender / Borrower Arbitration via Admin

- Lenders can report book condition damage after return.
- Admin can view complaint details and decide on the deposit deduction percentage.
- Users receive an in-app notification and can claim their deposit from the "My Deposit" page once arbitration is complete.

Admin Dashboard

- Implemented email notification to admin when a complaint or issue requires admin review.
- Further UI enhancements to the admin dashboard.
- Improved order issue handling from the admin side.

Production Deployment

- Deployed the latest build to the VPS production server.

Sprint 6 — Week ending 12 May 2026

Focus: Payment security documentation and final Stripe architecture clarification.

Payment Security — Architecture Overview

- The platform does not store any card or bank information. All card processing is handled directly by Stripe; the platform only stores a PaymentIntent ID per order.

- Lenders set up their payout account via Stripe's own onboarding page (Connect); the platform stores only their Stripe Account ID.

Deposit Refund Integrity

The mechanism ensuring the correct borrower receives their deposit refund:

- When a borrower pays, Stripe generates a unique PaymentIntent ID which is immediately stored in the database, linked to the specific borrower and order.
- When a refund is triggered, the platform retrieves the stored PaymentIntent ID and passes it to Stripe's refund API, which automatically returns the funds to the original payment card.
- This ensures refund accuracy regardless of how much time has passed since the original payment.

Damage Deduction Calculation

- Deduction amounts are calculated by the platform backend (not by Stripe).
- When admin confirms a damage deduction, the backend calculates the exact split (e.g., 25% to lender, 75% refunded to borrower) and sends those specific amounts to Stripe to execute.

3. Feature Completion Summary

Feature / Module	Status	Notes
User registration & email OTP verification	Completed	Brevo / bookborrow.org
Forgot / reset password via email	Completed	Reactivated from Phase 1 codebase
Change password (profile page)	Completed	—
Book listing (lender view)	Completed	Service fee dropdown, deposit tooltip, rental/day slider
Book search & filter	Completed	Mobile-responsive filter panel
Checkout & rental fee calculation	Completed	Rental/day × days; fixed \$2 platform fee; auto shipping calc
Stripe payment integration	Completed	PaymentIntent-based; sandbox tested
Shipping workflow	Completed	Manual tracking number; hold-then-release payment logic
Shipment email notification	Completed	Tracking number emailed to buyer on dispatch
Deposit hold & release	Completed	Held until return; released via Stripe refund API
Damage reporting & deposit deduction	Completed	Lender reports; admin arbitrates; split executed by backend
Borrower deposit claim ("My Deposit")	Completed	Post-arbitration claim flow

Feature / Module	Status	Notes
Refund management (role-based)	Completed	User view vs. admin approve/reject view
Complaint submission & management	Completed	Order ID dropdown; admin can reject or resolve
Admin dashboard — User / Book / Order metrics	Completed	KPI cards, filterable tables
Admin dashboard — Financial metrics	Completed	Fees, volume, refund rate, top earners
Admin dashboard — Shipping dashboard	Completed	Tracking status, missing numbers, shipment history
Admin platform fee setting (dynamic)	Completed	DB-stored; updated via admin panel API
Admin email notification (review required)	Completed	—
Admin authentication security fix	Completed	Boolean flag; no longer string-matched
Mobile responsive design	Completed	Key pages refactored; chat layout improved
Brevo email infrastructure	Completed	Team-owned account; DNS authenticated on Namecheap
VPS deployment (production)	Completed	Latest build live on production server

4. Known Issues & Handover Notes

The following items should be noted by the client and any future maintainers:

Item	Detail
Stripe — Production keys	Stripe is currently configured in sandbox mode. Production API keys must be configured before the platform goes live.
Admin account	A real production admin account must be set up in the live environment.
Deposit split rules	Damage deposit splits are currently decided manually by the admin. There are no automated rules — this is a candidate for future automation.
Brevo / DNS	DNS and CNAME records for bookborrow.org are configured via Namecheap. Admin rights are held by Atif. These records must remain active for email to function.
VPS access	Full VPS credentials should be formally transferred to the client or next maintainer. Contact Kevin or Atif for current access details.

5. Recommendations for Future Development

- **Stripe Production Migration:** Switch from sandbox to live Stripe API keys and complete merchant onboarding.
- **Automated Deposit Split Rules:** Replace manual admin arbitration with configurable rules (e.g., percentage scales based on damage severity).
- **Real-time Shipment Tracking:** Integrate with any possible Post API to provide live tracking status instead of manual tracking number entry.
- **Book Condition History:** Implement a full condition history log per book, recording assessments from both lender and borrower over multiple rental cycles.
- **Automated End-to-End Testing:** Build a test suite covering critical payment, refund, and complaint flows.
- **Enhanced Admin Analytics:** Extend the financial dashboard with trend charts, user retention metrics, and exportable reports.